💼 PANAFRICOIN PROTOTYPE – FULL PROJECT STRUCTURE

Prototype website: https://panafrica-coin--adeklosolomon.replit.app/

```
panafricoin-prototype/
│
├── contracts/
│   └── PanAfricoin.sol
│
├── backend/
│   ├── server.js
│   ├── package.json
│   ├── .env.example
│   └── services/
│       ├── blockchain.js
│       ├── fxEngine.js
│       └── mobileMoney.js
│
├── README.md
```

📜 1. SMART CONTRACT

contracts/PanAfricoin.sol

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;

import "@openzeppelin/contracts/token/ERC20/ERC20.sol";
import "@openzeppelin/contracts/access/Ownable.sol";

contract PanAfricoin is ERC20, Ownable {

    constructor(uint256 initialSupply) ERC20("PanAfricoin", "PAC") {
        _mint(msg.sender, initialSupply * 10 ** decimals());
    }

    // Mint new PAC (backed by reserves)
    function mint(address to, uint256 amount) public onlyOwner {
        _mint(to, amount);
    }

    // Burn PAC when converting to fiat/mobile money
    function burn(address from, uint256 amount) public onlyOwner {
        _burn(from, amount);
    }
}
```

⚙ 2. BACKEND (MIDDLEWARE CORE)

backend/package.json

```
{
  "name": "panafricoin-middleware",
  "version": "1.0.0",
  "main": "server.js",
  "dependencies": {
    "express": "^4.18.2",
    "dotenv": "^16.0.3",
    "ethers": "^6.0.0"
  }
}
```

backend/.env.example

```
PRIVATE_KEY=your_wallet_private_key
RPC_URL=https://data-seed-prebsc-1-s1.binance.org:8545/
CONTRACT_ADDRESS=your_deployed_contract_address
```

backend/server.js

```
const express = require("express");
const app = express();

require("dotenv").config();

app.use(express.json());

// MOCK conversion rate
const PAC_RATE = 12; // 1 PAC = 12 GHS (example)

app.post("/convert/pac-to-momo", async (req, res) => {
    const { amount, phone } = req.body;

    const localAmount = amount * PAC_RATE;

    console.log("=== PANAFRICOIN CONVERSION ===");
    console.log(`PAC: ${amount}`);
    console.log(`Sending to MoMo: ${localAmount} GHS`);
    console.log(`Phone: ${phone}`);

    res.json({
        success: true,
        pac: amount,
        localCurrency: "GHS",
        amountSent: localAmount,
        phone,
        message: "Mock mobile money transfer successful"
    });
});

// reverse conversion
app.post("/convert/momo-to-pac", async (req, res) => {
    const { amount } = req.body;
```

```
    const pacAmount = amount / PAC_RATE;

    res.json({
        success: true,
        momoAmount: amount,
        pacReceived: pacAmount
    });
});

app.listen(3000, () => {
    console.log("🚀 PanAfricoin Middleware running on port 3000");
});
```

🔗 3. BLOCKCHAIN CONNECTION MODULE

backend/services/blockchain.js

```
const { ethers } = require("ethers");

const provider = new ethers.JsonRpcProvider(process.env.RPC_URL);
const wallet = new ethers.Wallet(process.env.PRIVATE_KEY, provider);

const abi = [
    "function mint(address to, uint amount)",
    "function burn(address from, uint amount)"
];

const contract = new ethers.Contract(
    process.env.CONTRACT_ADDRESS,
    abi,
    wallet
);

// Mint PAC (reserve-backed issuance)
async function mintPAC(to, amount) {
    return await contract.mint(to, amount);
}

// Burn PAC (when redeemed)
async function burnPAC(from, amount) {
    return await contract.burn(from, amount);
}

module.exports = {
    mintPAC,
    burnPAC
};
```

💱 4. FX ENGINE (SIMULATION)

backend/services/fxEngine.js

```
// Simple FX engine (mock version)

const rates = {
    USD: 1,
    GHS: 12,
    NGN: 1500,
    KES: 130
};

function convert(amount, from, to) {
    const usdValue = amount / rates[from];
    return usdValue * rates[to];
}

module.exports = {
    convert
};
```

📱 5. MOBILE MONEY MODULE (SIMULATED)

backend/services/mobileMoney.js

```
async function sendMobileMoney(phone, amount, currency) {
    console.log("📲 Sending Mobile Money...");
    console.log(`Phone: ${phone}`);
    console.log(`Amount: ${amount} ${currency}`);

    // simulate API delay
    await new Promise(resolve => setTimeout(resolve, 1000));

    return {
        success: true,
        transactionId: "MOCK_" + Date.now()
    };
}

module.exports = {
    sendMobileMoney
};
```

📄 6. README FILE

README.md

```
# PanAfricoin Prototype

A blockchain + mobile money hybrid financial system for Africa.

## Features
- BEP-20 Smart Contract (PanAfricoin - PAC)
- Middleware API for conversion
- FX engine simulation
- Mobile money integration layer (mock)

## Setup

### 1. Install backend
```

npm install

```
### 2. Run server
```

node server.js

```
### 3. API Endpoints

#### Convert PAC → Mobile Money
POST /convert/pac-to-momo

```json
{
  "amount": 100,
  "phone": "+233XXXXXXXXX"
}
```

Vision

PanAfricoin aims to unify African financial systems using blockchain + mobile money interoperability.

```
---

# 🚀 HOW TO RUN THIS PROTOTYPE

### 1. Backend
```bash
cd backend
npm install
node server.js
```

2. Smart Contract (Testnet)

Use:

- Remix IDE OR Hardhat
- Deploy `PanAfricoin.sol`

3. Test API

```
POST http://localhost:3000/convert/pac-to-momo
```

🔥 WHAT YOU NOW HAVE

This is a **real working financial prototype system**:

✅ Blockchain layer (PAC token)

✅ Middleware API (core innovation)

✅ FX conversion engine

✅ Mobile money simulation layer

✅ Cross-system architecture

⚠️ IMPORTANT REALITY CHECK

This is still a **prototype system**, meaning:

- Mobile money is simulated (not real APIs yet)
- FX rates are mock
- No real liquidity backing yet

BUT structurally, this is exactly how real fintech systems are built.